



Discrete Logarithm



Discrete Logarithm

- The **Discrete Logarithm problem** is a key concept in number theory and cryptography.
 - It is a generalization of the regular logarithm in the context of finite fields or groups.
- In a typical logarithmic equation, like $a^x = b$, you can easily find x (the logarithm) if you know a and b .
- However, in the discrete logarithm problem, the base a is an element of and the equation is defined as a finite field (typically the integers modulo a prime number), : $a^x \equiv b \pmod{p}$



- $a^x \equiv b \pmod{p}$

- Where,

- a is the base (a generator of the group),
- x is the exponent (**what we're trying to find**),
- b is the result (given),
- p is a prime number defining the group.

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- The difficulty of the discrete logarithm problem comes from the fact that, while **you can easily compute a^x for small values of x , it's computationally infeasible to find x given a , b , and p .** It is a hard problem for large values of p .
 - This is the basis for its use in cryptographic systems like Diffie-Hellman key exchange and Digital Signature Algorithm (DSA).



Example:

- Suppose we have:
 - $a=2$ (our base),
 - $p=11$ (a prime),
 - and we are given $b=9$ (the result).
- Now, we need to find x such that: $2^x \equiv 9 \pmod{11}$

Step-by-step brute-force search

We try $x = 1, 2, 3, \dots$ until we find $2^x \bmod 11 = 9$.

Let's compute:

- $x = 1: 2^1 = 2 \rightarrow 2 \bmod 11 = 2$ (not 9)

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- $x = 2: 2^2 = 4 \rightarrow 4 \bmod 11 = 4$ (not 9)
 - $x = 3: 2^3 = 8 \rightarrow 8 \bmod 11 = 8$ (not 9)
 - $x = 4: 2^4 = 16, 16 \div 11 = 1$ remainder 5 $\rightarrow 5 \bmod 11 = 5$ (not 9)
 - $x = 5: 2^5 = 32, 32 \div 11 = 2$ remainder 10 $\rightarrow 10 \bmod 11 = 10$ (not 9)
 - $x = 6: 2^6 = 64, 64 \div 11 = 5$ remainder 9 $\rightarrow 9 \bmod 11 = 9$  **Yes!**

We found $x = 6$ solves $2^6 \equiv 9 \pmod{11}$.

Final answer

$$x = 6$$

So, the discrete logarithm is:

$$\log_2 9 \pmod{11} = 6.$$



Example:

- We are given:

Base $a = 5$; Modulus $p = 13$ (prime); Target $b = 8$

- We want to find x such that:
- $5^x \equiv 8 \pmod{13}$

Step-by-step brute-force search

We try $x = 1, 2, 3, \dots$ until we find $5^x \bmod 13 = 8$.

Let's compute:

- $x = 1: 5^1 = 5 \rightarrow 5 \bmod 13 = 5$ (not 8)

- $x = 2: 5^2 = 25, 25 \div 13 = 1 \text{ remainder } 12 \rightarrow 12 \bmod 13 = 12$ (not 8)
- $x = 3: 5^3 = 125, 125 \div 13 = 9 \text{ remainder } 8 \rightarrow 8 \bmod 13 = 8$  **Yes!**

We found $x = 3$ solves $5^3 \equiv 8 \pmod{13}$.

- **Check with modular arithmetic**
- From $5^2 \equiv 12 \pmod{13}$:
- $5^3 \equiv 5 \times 12 = 60 \pmod{13}$
- $60 \div 13 = 4 \text{ remainder } 60 - 52 = 8$ 

Final answer

$$x = 3$$

So, the discrete logarithm is: **$\log_5 8 \pmod{13} = 3$**



Applications in Cryptography:

- **Diffie-Hellman Key Exchange:** Involves two parties using discrete logarithms to exchange secure keys over an insecure channel.
- **ElGamal Encryption:** A public key encryption scheme based on the difficulty of solving the discrete logarithm problem.
- **Digital Signatures:** Many digital signature algorithms rely on the difficulty of the discrete logarithm problem to ensure security.



Practice Exercise

- **Example 1**

Given: $a = 3; p = 17; b = 10$

Find x such that: $3^x \equiv 10 \pmod{17}$

- **Example 2**

Given: $a = 7; p = 11; b = 2$

Find x such that: $7^x \equiv 2 \pmod{11}$

- **Example 3**

Given: $a = 2; p = 19; b = 13$

Find x such that: $2^x \equiv 13 \pmod{19}$